

ABDUL AZIZ

Lahore, Pakistan

+92 343 4489592 | mughalabdulaziz4@gmail.com | [linkedin.com/in/abdul-aziz-a6670a249](https://www.linkedin.com/in/abdul-aziz-a6670a249) | github.com/mindbruster

SUMMARY

AI Engineer specialising in production LLM systems — RAG and vector retrieval, multi-model orchestration, and content-safety guardrails — built as complete products rather than prototypes, with the payments, real-time dashboards, and multi-tenant APIs around them. Took five platforms to production in a single year across children's education, legal technology, generative art, AI knowledge management, and AI-cost attribution, including safety guardrails filtering every model response for an under-13 audience in 20+ languages. Earlier background in computer vision (YOLOv8, ANPR, OCR) deployed with a government surveillance partner.

EXPERIENCE

BIG Immersive

Associate GenAI Engineer

Aug 2025 – Present

Nov 2025 – Present

- Ship production GenAI features end to end across **five client and internal platforms** — children's education (Yoluno), legal technology (Graysen), generative art (SoundBYTE), AI knowledge management (myNeutron), and AI-cost attribution (LLM Meter) — owning work from model integration through deployment.
- Build **RAG pipelines and vector search** over production knowledge bases, including the semantic memory and retrieval modules powering myNeutron and citation-grounded legal retrieval over UAE federal law.
- Orchestrate **multiple LLM providers** — OpenAI, Gemini via OpenRouter, and Replicate — behind a single service layer spanning chat, vision, transcription, image generation, and TTS, with per-call timeouts protecting API workers.
- Design and implement **LLM content-safety and moderation layers**, including a multi-stage filter that classifies every model response before it reaches a child user.
- Deliver the surrounding product surface: **multi-tenant APIs**, Stripe billing, real-time dashboards over Socket.io, and Dockerised services deployed on Railway.

AI Excellence Program

Aug 2025 – Oct 2025

- Built a **RAG-based chatbot** delivering context-aware responses grounded in a custom document corpus.
- Contributed to AI/ML projects across model building, evaluation, and deployment, and collaborated with the team on live client work.

FYP Collaboration — Punjab Safe Cities Authority (PSCA)

Nov 2024 – Jan 2025

AI Engineer (Final Year Project)

Lahore, Pakistan

- Built a **multi-stage computer vision pipeline** for automated litter-offence detection: **YOLOv8s** for littering-action detection, **YOLOv8n** for offending-vehicle tracking, then **ANPR** and **OCR** to extract plate numbers and generate challans automatically.
- Scraped and curated a custom **30,000+ image dataset**, fine-tuned both YOLO models, and validated real-time performance against live road footage with PSCA's operations team.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C/C++, SQL, HTML/CSS

GenAI & LLM: RAG, vector search, embeddings, semantic memory, prompt engineering, multi-model orchestration (OpenRouter), content moderation & safety guardrails, LangChain, OpenAI, Gemini, Anthropic, Replicate

ML & Computer Vision: TensorFlow, PyTorch, scikit-learn, NumPy, Pandas, OpenCV, YOLOv8, ANPR, OCR, dataset curation & fine-tuning

Backend: FastAPI, Node.js, Express, Next.js, REST APIs, Prisma, Socket.io, JWT/OAuth, Stripe, Docker

Frontend: React, Next.js, SvelteKit, Vite, Tailwind CSS, Radix/shadcn, TanStack Query, Zustand

Data & Infrastructure: PostgreSQL, MongoDB, MySQL, Redis, Firebase, AWS S3, Railway, Vercel

Practices: Multi-tenant architecture, rate limiting, webhooks, E2E testing (Playwright), Git

PROJECTS

Yoluno | *React, TypeScript, Express, PostgreSQL, Gemini via OpenRouter, Socket.io, Stripe*

yoluno.com

- Child-safe AI learning platform for ages 3–13: age-adaptive AI companions, illustrated story generation, and family-history features across 20+ languages, grounded in the English National Curriculum and IB (KS1–KS4).
- Built a **three-layer content-safety pipeline** — banned-topic matching, keyword expansion, and typo-tolerant fuzzy matching — classifying every message green/yellow/red and escalating flagged content to parents in real time over Socket.io.
- Orchestrated **multiple Gemini models via OpenRouter** for chat, vision, story text, illustration, and transcription, with OpenAI TTS for narration and per-call abort timeouts protecting the API workers.

Graysen | RAG, LLMs, Multilingual NLP

graysen.ai

- AI contract analyzer for UAE law that flags problematic clauses — hidden renewals, excessive penalties, ambiguous terms — and returns a contract health score, with AI-suggested rewrites as negotiation starting points.
- Built a legal chat **grounded in UAE federal law with source citations**, plus a templated document builder, across both English and Arabic contracts.

SoundBYTE® / Sonic Identity™ | React monorepo, Express, Prisma, Replicate, Gemini, Stripe

amritasethi.com

- Voice-to-art generative platform for Dubai's first NFT artist (featured on CNN, Forbes, BBC): captures a voice recording in-browser via MediaRecorder and transforms it into a unique generative visual identity.
- Shipped a **3-app npm-workspaces monorepo** (marketing site, creation playground, admin console) over a shared Express + Prisma API with Stripe payments, Redis sessions, S3 storage, and print-on-demand fulfilment.

myNeutron | RAG, vector search, semantic memory, embeddings, multi-model LLM orchestration

myneutron.ai

- AI knowledge base that turns scattered browsing, files, and chat history into reusable context: users save *Seeds* (pages, files, notes, AI chats), which are auto-organised into *Bundles*, queried in natural language, and fed back into ChatGPT, Claude, and other assistants — with an optional on-chain storage tier.
- Built the **semantic memory and retrieval modules** — chunking and embedding of mixed-format sources, **vector search** over each user's corpus, and context-aware bundling that surfaces related material across the knowledge base rather than one document at a time.

LLM Meter | Next.js 15, Prisma, PostgreSQL, Recharts, Multi-tenant

- Cross-provider AI spend **attribution** platform answering which team, project, or person spent what on AI — tracking OpenAI, Anthropic, Gemini, OpenRouter, fal.ai, and GLM in one place.
- Built API-key-authenticated multi-tenant ingestion with daily rollups, and a **subscription seat-cost allocation model** that attributes flat-fee AI seats down to per-project cost.

Sales Pitch Generator | SvelteKit, FastAPI, Gemini API, RAG

- AI platform that automatically generates sales pitch presentations, using RAG over company material and Gemini for contextual content generation.

Authentication App with ChatBot | SvelteKit, PostgreSQL, RAG, OAuth

- Chatbot web app with editable prompts, branching conversation history, OAuth login, and admin controls; integrated RAG for dynamic knowledge retrieval from stored conversation history.

EDUCATION

FAST NUCES, Lahore

Aug 2021 – July 2025

Bachelor of Science in Computer Science

Lahore, Pakistan

- **Relevant Coursework:** Artificial Intelligence, Machine Learning, Computer Vision, Data Science, Data Structures & Algorithms, Database Systems, Operating Systems, Computer Networks.

ACHIEVEMENTS

- Shipped **five production AI platforms** within the first year as an engineer — spanning children's education, legal technology, generative art, AI knowledge management, and internal cost tooling.
- Designed and shipped **LLM safety guardrails for a children's product**, classifying every model interaction across three detection layers before it reaches a user aged 3–13, in 20+ languages.
- Built a **subscription seat-cost attribution model** that allocates flat-fee AI seats down to per-project cost — a capability not available in any off-the-shelf AI spend tool.
- Curated a **30,000+ image training dataset** and fine-tuned YOLOv8 detection and tracking models, achieving real-time performance on live road footage for a government surveillance deployment.